Problem-Solution Fit

Connecting user pain to product capability

Project: AI WeatherWise API   Track: AI-ML & GEN-AI Track

Prepared: 07 October 2026   Team ID: Not provided

Maximum Marks: 3

Team Members & Assigned Roles

Team Member	Role
Jai Bharath A	Team Lead & Lead Backend/API Architect
Mukesh Balaji	Generative AI & Prompt Engineering Engineer
Yadagiri Mani	Database & Data Engineering Specialist
Deepak G	Security, Testing & Reliability Engineer
Naveen	API Integration, Documentation & Demonstration Engineer

User pain	Solution mechanism	Expected value	Fit
Weather numbers are hard to interpret	Gemini natural-language summary	Quick understanding	Strong
No practical action guidance	AI recommendation engine	Better daily decisions	Strong
Frequent cities require repeated input	Saved location CRUD	Convenience	Strong
External API outages	Weather mock fallback	Resilience	Strong
AI quota/network failures	Rule-based fallback	Continuity	Strong
Sensitive user data	JWT + bcrypt + ownership guard	Security	Strong

Differentiation

The project is not another weather data proxy. Its core value is the intelligence layer that converts measurements into decisions while preserving a resilient engineering path when cloud dependencies fail.

Fit Risks

- Recommendations can become generic if prompts are not grounded in current weather context.
- Free-tier AI quotas can constrain high-volume testing.
- Mock data must be clearly distinguishable from live upstream observations.
- Personalization is limited until the API captures richer user preferences.